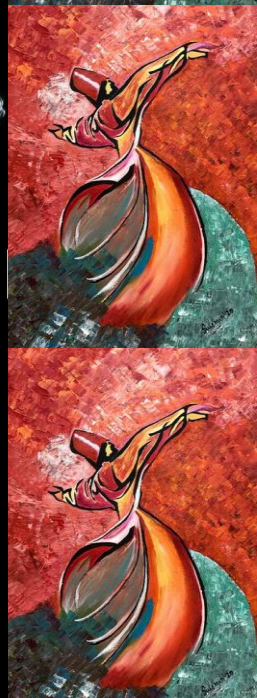




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



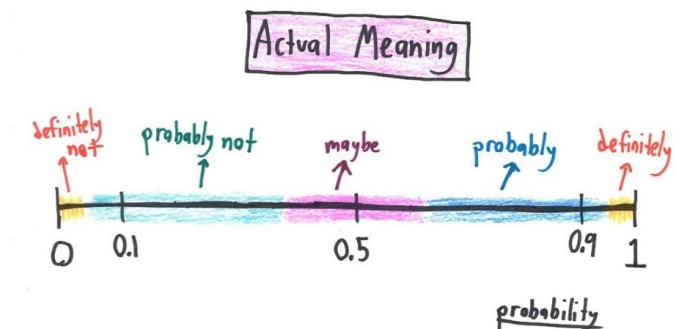


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 10 - Lecture No 19 – October 22, 2025



Announcements!

1. All lectures posted.
2. Scores for all 5 quizzes and MT posted.
3. **Quiz No 06** – Monday October 27, 2025 (DRV – Conditional probability and Total Probability)

Agenda for today

1. Quiz No 07
2. Complete Unit 4 - Discrete Random Variables

Independence and Expectations

- In general: $\mathbf{E}[g(X, Y)] \neq g(\mathbf{E}[X], \mathbf{E}[Y])$ *always true*
- Exceptions: $\mathbf{E}[aX + b] = a\mathbf{E}[X] + b$ $\mathbf{E}[X + Y + Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$

If X, Y are **independent**:

$$\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]$$

$g(X)$ and $h(Y)$ are also independent:

$$\mathbf{E}[g(X) h(Y)] = \mathbf{E}[g(X)] \mathbf{E}[h(Y)]$$

$$\mathbf{E}[g(X, Y)]$$

$$g(x, y) = xy$$

$$\rightarrow = \sum_x \sum_y (xy) p_{X,Y}(x,y) = \sum_x \sum_y (xy) p_X(x) p_Y(y)$$

$$\rightarrow = \sum_x x p_X(x) \sum_y y p_Y(y) = \mathbf{E}[X] \mathbf{E}[Y]$$

Independence and Variances

- Always true: $\text{var}(aX + b) = a^2 \text{var}(X)$ $\text{var}(X + a) = \text{var}(X)$
- In general: $\text{var}(X + Y) \neq \text{var}(X) + \text{var}(Y)$

If X, Y are **independent**: $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

assume

$$E[X] = 0 ; E[Y] = 0$$

$$E[X]E[Y] = 0$$

$$\begin{aligned} \Rightarrow \text{var}(X + Y) &= E[(X + Y)^2] = E[X^2 + 2XY + Y^2] \\ \Rightarrow &= E[X^2] + 2E[XY] + E[Y^2] = E[X^2] + E[Y^2] = \text{var}(X) + \text{var}(Y) \end{aligned}$$

- Examples:

➤ If $X = Y$: $\text{var}(X + Y) = \text{var}(2X) = 4\text{var}(X)$

➤ If $X = -Y$: $\text{var}(X + Y) = \text{var}(0) = 0$

➤ If X, Y independent: $\text{var}(X - 3Y) = \text{var}(X) + \text{var}(-3Y) = \text{var}(X) + 9\text{var}(Y)$

Variance of the binomial

- X : binomial with parameters n, p
 - ❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p
(indicator variable)
 $X_i = 0$ otherwise $1-p$

independent

$$X = X_1 + \dots + X_n$$

$$\Rightarrow \text{var}(X) = \text{var}(X_1) + \text{var}(X_2) + \dots + \text{var}(X_n)$$

$$\Rightarrow \text{var}(X) = n \text{var}(X_1) = np(1-p)$$

The variance of the geometric PMF

X: geometric parameter with p

- Given: $X > 1$, $X - 1$ has the same geometric PMF (unconditional PMF of X)
- Given $X > 1$, conditional PMF of X: same as unconditional PMF of $X + 1$

$$\rightarrow E[X \mid X > 1] = 1 + E[X] \quad E[X^2 \mid X > 1] = E[(X + 1)^2]$$

$$\begin{aligned} \rightarrow E[X^2] &= P(X=1) E[X^2 \mid X=1] + P(X > 1) E[X^2 \mid X > 1] \\ &= p \cdot 1 + (1-p) (E[X^2] + 2E[X] + 1) \end{aligned}$$

$\xrightarrow{\hspace{10em}} 1/p$

$$\text{Algebra: } E[X^2] = \frac{2}{p^2} - \frac{1}{p}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Poisson RV

Parameter: λ (a positive real number characterizing the average number of events in the interval)

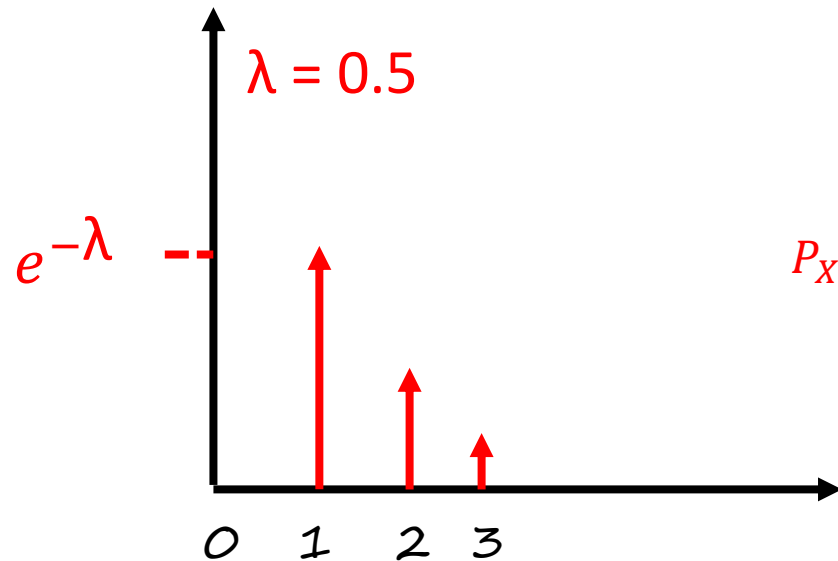
$$P_X(k) = e^{-\lambda} \frac{\lambda^K}{K!}, K = 0, 1, 2, 3, \dots$$

Models: Probability of occurrence of events in a fixed interval of time or space if these events occur with a known constant rate and independent of time of last event.

Note:
$$\sum_{K=0}^{\infty} e^{-\lambda} \frac{\lambda^K}{K!} = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} \dots \right)$$
$$= e^{-\lambda} * e^{\lambda} = 1$$

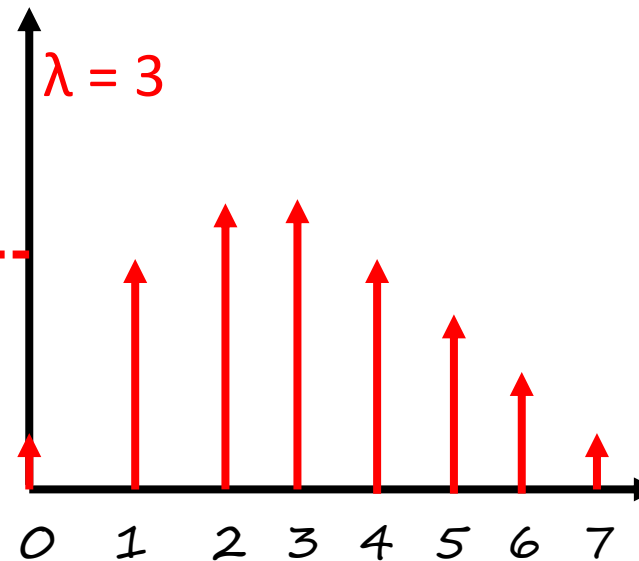
Recognize the Taylor series: The remaining sum, $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$, is the well-known Taylor series expansion of e^x evaluated at $x = \lambda$. The general form is:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots$$



Note: When $\lambda < 1$, PMF, is monotonically decreasing with K

$$P_X(K) = e^{-\lambda} \frac{\lambda^K}{K!}$$



Note: When $\lambda > 1$, PMF first increases and then decreases

Poisson RV: Example

$$P_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

An IC manufacturing company provides 100 IC's per day. Defective IC's are at a fixed rate of 3 IC's/day. Probability of R defective IC's in a given day.

$$P_X(0) = e^{-3} \approx 0.05$$

$$P_X(3) = e^{-3} \frac{3^3}{3!} = 0.05 * 4.5 = 0.2256$$

How do you find $P(X > 3)$?

Useful in Modeling

- a) No. of phone calls received by a call center (per hour, per day, or per anytime unit)
- b) No. of photons hitting a detector in a particular time interval

Discrete Random Variables - Summary

- r.v.'s and PMFs: $p_X(x), p_{X,Y}, p_{X|Y}(x|y), p_{X|A}(x)$
- Expectation: $E[X], E[X|A], E[X|Y = y]$
Expected value rule: $E[g(X,Y)], E[g(X,Y)|A], E[g(X,Y) | Z = z]$
Linearity: $E[aX + bY] = aE[X] + bE[Y]$
- Variance: $\text{var}(X), \text{var}(X|A), \text{var}(X|Y = y)$
 $\text{var}(X) = E[X^2] - (E[X])^2$
- Independence of r.v.'s: $p_{X,Y} = p_X * p_Y$
 $E[XY] = E[X] * E[Y]$
 $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$
- Multiplication Rule
 $p_{X,Y,Z}(x, y, z) = p_Z(z) p_{Y|Z}(y|z) p_{X|Y,Z}(x|y, z)$
- Total Probability Theorem
 $p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y)$
- Total expectation Theorem
 $E[X] = \sum_y p_Y(y) E[X|Y = y]$
- Examples: Bernoulli, indicators, uniform, binomial, geometric

Congrats!

End of Unit 4!

End of Chapter Exercise - The hat problem

- n people throw their hats in a box and then pick one at random
 - ❖ All permutations equally likely $1/n!$
 - ❖ Equivalent to picking one hat at a time

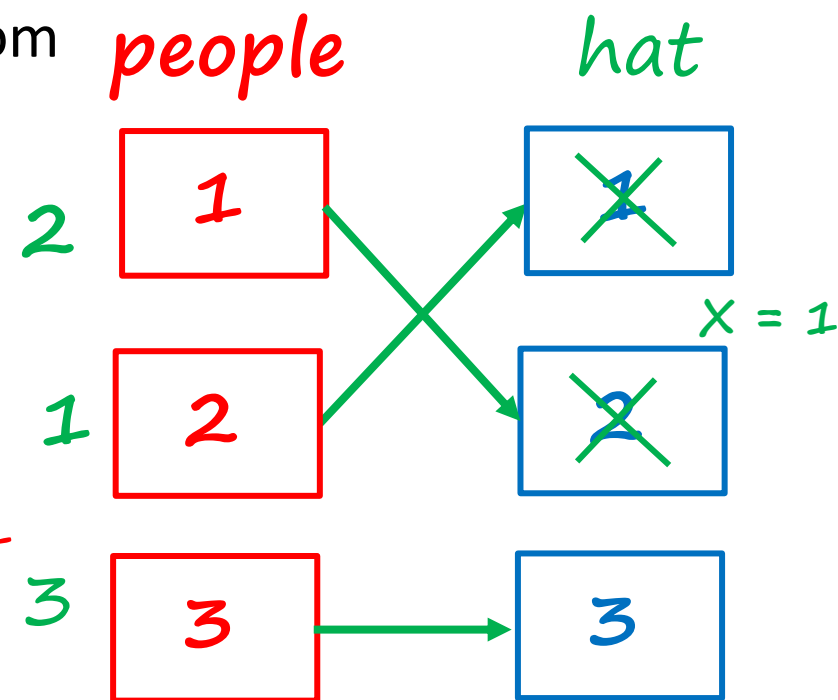
- X : number of people who get their own hat

❖ Find $E[X] = E[X_1] + \dots + E[X_n] = n * 1/n = 1$

$$X_i = \begin{cases} 1, & \text{if } i \text{ selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$$X = X_1 + \dots + X_n$$

❖ $E[X_i] = E[X_1] = P(X_1 = 1) = 1/n$



$$\frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{3!}$$

$P_X(k)$: hard!

$$\sum_k k p_X(x)$$

The variance in the hat problem

- X : number of people who get their own hat
❖ Find $\text{var}(X)$

$$X_i = \begin{cases} 1, & \text{if } i \text{ selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$$\bullet \text{ } \boxed{\text{var}(X)} = E[X^2] - (E[X])^2 = 2 - 1 = \boxed{1}$$

$$\rightarrow \bullet E[X_i^2] = E[X_1^2] = E[X_1] = 1/n \quad \rightarrow E[X^2] = n \frac{1}{n} + n(n-1) \frac{1}{n} * \frac{1}{n-1}$$

$$\rightarrow \bullet \text{ for } i \neq j: E[X_i X_j] = E[X_1 X_2] = P(X_1 X_2 = 1) = P(X_1 = 1, X_2 = 1) \\ \rightarrow = P(X_1 = 1) \cdot P(X_2 = 1 \mid X_1 = 1) = \frac{1}{n} * \frac{1}{n-1}$$

$$n = 2$$

$$X_1 = 1 \Rightarrow X_2 = 1$$

$$X_1 = 0 \Rightarrow X_2 = 0$$

$$X = X_1 + \dots + X_n$$

$$X^2 = \sum_i^n X_i^2 + \sum_{i,j: i \neq j}^{n(n-1)} X_i X_j$$

Question 1 - (15 points)

Let's be optimistic and assume that Pakistan cricket team has qualified for the T20 World Cup Semi Final. Pakistan's opponent in Semi Final will be West Indies, England, or Australia with a probability of 0.4, 0.4, and 0.2 respectively. The probability of Pakistan winning against West Indies is 0.6. The probability of Pakistan winning against England is 0.4. The probability of Pakistan winning against Australia is 0.6.

- a) (7.5 points) What is the probability that Pakistan will win the Semi-final.
- b) (7.5 points) Assume that you decided to completely switch off from cricket and did not follow the game. Later, you were told that Pakistan lost the Semi Final. What is the probability that Pakistan played the Semi Final game against England?

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Solution:

(a)

Let's assume the following naming conventions for various events:

$A_1 = \{ \text{Pakistan play West Indies in Semi Final} \}$

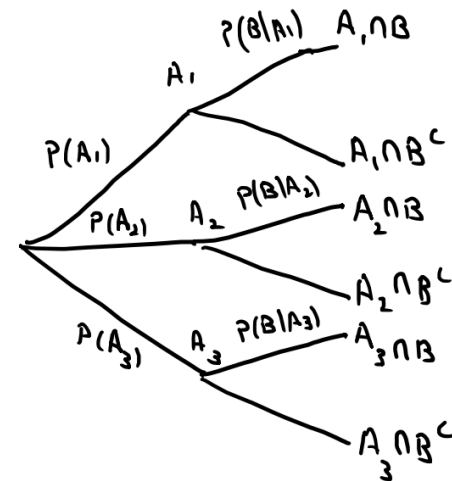
$A_2 = \{ \text{Pakistan play England in Semi Final} \}$

$A_3 = \{ \text{Pakistan play Australia in Semi Final} \}$

$B = \{ \text{Pakistan win the Semi Final} \}$

$B^c = \{ \text{Pakistan lose the Semi Final} \}$

$$\Rightarrow \begin{array}{lll} P(A_1) = 0.4 & P(A_2) = 0.4 & P(A_3) = 0.2 \\ P(B|A_1) = 0.6 & P(B|A_2) = 0.4 & P(B|A_3) = 0.6 \\ P(B^c|A_1) = 0.4 & P(B^c|A_2) = 0.6 & P(B^c|A_3) = 0.4 \end{array}$$



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By Total Probability Theorem:-

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= (0.4)(0.6) + (0.4)(0.4) + (0.2)(0.6)$$

$$= 0.24 + 0.16 + 0.12$$

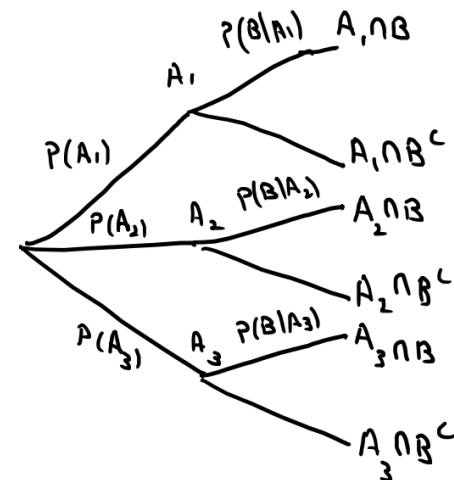
$$P(B) = 0.52 \quad \text{And}$$

$$P(B^c) = 1 - P(B) = 0.48$$

$$\Rightarrow P(A_1) = 0.4 \quad P(A_2) = 0.4 \quad P(A_3) = 0.2$$

$$P(B|A_1) = 0.6 \quad P(B|A_2) = 0.4 \quad P(B|A_3) = 0.6$$

$$P(B^c|A_1) = 0.4 \quad P(B^c|A_2) = 0.6 \quad P(B^c|A_3) = 0.4$$



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$$P(B^c) = 1 - P(B) = 0.48$$

According to Bayes' Rule,

$$P(A_2|B^c) = \frac{P(A_2)P(B^c|A_2)}{P(B^c)}$$

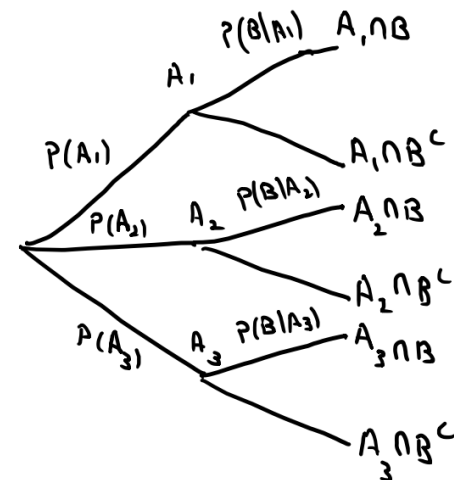
$$= \frac{(0.4)(0.6)}{0.48}$$

$$P(A_2|B^c) = 0.5 \quad \text{--- Ans (b)}$$

$$\Rightarrow P(A_1) = 0.4 \quad P(A_2) = 0.4 \quad P(A_3) = 0.2$$

$$P(B|A_1) = 0.6 \quad P(B|A_2) = 0.4 \quad P(B|A_3) = 0.6$$

$$P(B^c|A_1) = 0.4 \quad P(B^c|A_2) = 0.6 \quad P(B^c|A_3) = 0.4$$



Question 3 - (14 points)

Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

(If the first roll results in 1 and second roll results in 3, the value taken by X is 2)

(If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

Question 3 - (14 points)

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Y = 2X

- a) (4 points) Calculate and plot the PMF of X
- b) (4 points) Calculate and plot the PMF of Y
- c) (4 points) Calculate Mean of Y.
- d) (4 points) Calculate Variance of Y

Possible magnitudes of the difference between two rolls:

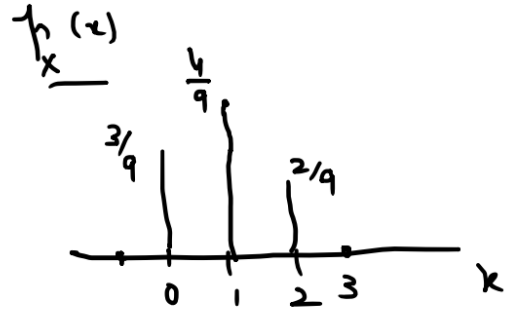
Roll 2 = 3	2	1	0
Roll 2 = 2	1	0	1
Roll 2 = 1	0	1	2
	Roll 1 = 1	Roll 1 = 2	Roll 1 = 3

Possible values that X can take = {0, 1, 2}

$P(X=0) = \frac{3}{9}$

$P(X=1) = \frac{4}{9}$

$P(X=2) = \frac{2}{9}$



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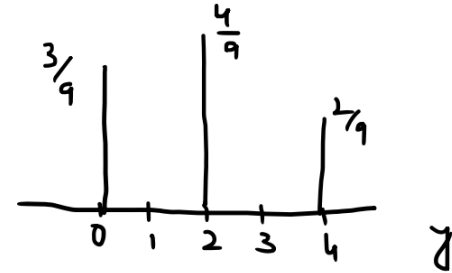
d) (4 points) Calculate Variance of Y

(b)

$$Y = 2X$$

Possible values of Y = $\{0, 2, 4\}$

$P_Y(y)$:



(c)

$$\begin{aligned} \mu_Y = E[Y] &= \sum_y y P_Y(y) \\ &= 0\left(\frac{3}{9}\right) + 2\left(\frac{4}{9}\right) + 4\left(\frac{2}{9}\right) \\ \mu_Y = E[Y] &= \frac{16}{9} = 0.17 \end{aligned}$$

Question 3 - (14 points)

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Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

(d)

$$\text{var}(Y) = E[Y^2] - (E[Y])^2$$
$$E[Y^2] = \sum y^2 p_Y(y) \quad (\text{Expected Value Rule})$$
$$= 0^2 \left(\frac{3}{9}\right) + 2^2 \left(\frac{4}{9}\right) + 4^2 \left(\frac{2}{9}\right)$$
$$= \frac{16}{9} + \frac{32}{9}$$
$$E[Y^2] = \frac{48}{9}$$
$$\text{var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2$$
$$\text{var}(Y) = 2.17$$

Question 4 - (4 points)

Consider two random variables X & Y:

X = Temperature in Karachi on Oct 29, 2021 in Celsius ($^{\circ}\text{C}$)

Y = Temperature in Karachi on Oct 29, 2021 in Fahrenheit ($^{\circ}\text{F}$)

Assume that you have been provided the PMF of X and the corresponding PMF of Y.

Case 1: You take the mean of X and then convert this mean value to $^{\circ}\text{F}$

Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$)

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Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$)

If Y is a linear function of X , i.e.

$$Y = aX + b$$

$$E[Y] = aE[X] + b$$

Here,

$$Y = 1.8X + 32$$

$$\Rightarrow E[Y] = 1.8E[X] + 32$$

Case 2

Computation in Case 1

Therefore, you will get the same result in both cases.